AF 52

Synopsis
“USB A+C CHARGER”

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering
In
Electronic & Computer Science Engineering

by

Shubham Chaugule [EU1237005]

Manish Devashi [EU1237016]

Halim Ansari [EU1237047]

Mushkaan Das [EU2237002]

Under the Guidance of
Shilpa Katre
Assistant Professor, ECS, SJCEM

DEPARTMENT OF ELECTRONIC & COMPUTER SCIENCE ENGINEERING
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI

2026-2027

St. John College of Engineering and Management
Autonomous Institute
(A Christian Religious Minority Institution)
Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE
DTE Code : 3218 AICTE Permanent ID : 1-4790201
NAAC Accredited with Grade 'A+', Three Programs NBA Accredited

Certificate of Approval

This is to certify that Shubham Chaugule , Manish Devashi , Halim Ansari , Mushkaan Das, are Bonafide students of Bachelor of Engineering in Electronic & Computer Science Engineering Department, St. John College of Engineering and Management, Mumbai. They have satisfactorily completed the requirements of Synopsis as prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on "USB A+C CHARGER".

Signature :
Name : Mrs. Shilpa Katre
Assistant Professor, ECS, SJCEM

Signature :
Name : Dr. Md. Imaduddin
Head of the Department of Electronic & Computer Science Engineering

Signature: ----------
Name : Dr. Kamal Shah
Principal,
St. John College of Engineering and Management.

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. we also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. we understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Student Name	Signature
Shubham Chaugule	
Manish Devashi	
Halim Ansari	
Mushkaan Das	

Acknowledgement

We express our sincere gratitude to all those who have directly and indirectly contributed to the successful preparation of this project synopsis.

We are deeply grateful to **Dr. Kamal Shah, Principal, St. John College of Engineering and Management**, for providing an excellent academic environment and the necessary facilities to undertake this project.

We extend our heartfelt thanks to the **Dr. Md. Imaduddin**, **Head of the Department of Electronics & Computer Science Engineering** for their constant encouragement, valuable suggestions, and continuous support throughout the preparation of this synopsis.

We would like to express our sincere appreciation to our project guide, **Prof. Shilpa Katre**, for their invaluable guidance, constructive suggestions, technical expertise, and continuous motivation. Their encouragement and insightful feedback have been instrumental in shaping this project synopsis.

We would also like to express our sincere gratitude to **Inensy** for providing us with the opportunity to gain valuable industry exposure and practical experience during the project. We extend our heartfelt thanks to our **Industry Mentor, Mr. Anand Shukla**, for his continuous guidance, valuable insights, technical support, and encouragement throughout the project. His expertise and constructive feedback have greatly contributed to our learning and understanding of industry practices.

We acknowledge all the faculty members and laboratory staff of the Department of Electronics & Computer Science Engineering for their timely assistance, cooperation, and technical support.

Finally, we express our heartfelt gratitude to our parents, family members, friends, and classmates for their constant encouragement, moral support, and cooperation throughout this work.

Group Members

1. **Shubham Chaugule**
2. **Manish Devashi**
3. **Halim Ansari**
4. **Mushkaan Das**

ABSTRACT

The **Automotive USB A+C Charger** is designed to provide a safe, reliable, and convenient charging solution for electronic devices inside vehicles. The system supports both **USB Type-A and USB Type-C** charging interfaces, allowing users to charge a wide range of devices such as smartphones, tablets, GPS devices, and other USB-powered electronics.

The charger is designed to operate from the vehicle's electrical supply and convert the input voltage into regulated output power suitable for connected devices. The system incorporates essential protection features such as **over-voltage protection, over-current protection, short-circuit protection, and thermal protection** to improve safety and reliability during operation.

The USB Type-C interface provides modern and efficient charging capability, while the USB Type-A interface maintains compatibility with conventional charging cables and devices. The project focuses on **power conversion, voltage regulation, protection mechanisms, thermal management, and compact automotive design**.

The proposed Automotive USB A+C Charger offers a practical and user-friendly charging solution for modern vehicles while maintaining compatibility with both legacy and newer USB charging standards.

C O N T E N T S

- List of Figures
- List of Tables
- Abbreviations and symbols

Chapter No.	Topic	Page No.
Chapter 1	Introduction: Background, Problem Statement, Need, Objectives, Scope, Expected Outcomes	**3-7**
Chapter 2	Literature Review: Survey table, Research Gap, Novelty/Innovation	**8-12**
Chapter 3	Proposed Methodology: Working Principle, Architecture/Block Diagram, Flowchart, Methodology, Components	**13-18**
Chapter 4	Project Planning: WBS, Gantt Chart, Roles, Risk Assessment	**19-21**
Chapter 5	Resources & Feasibility: Technical/Economic/Operational Feasibility, Budget, BOM	**22-23**
Chapter 6	Expected Results & Impact: Expected outcomes, Industrial relevance, Sustainability, SDG mapping, Future Scope	**24-26**
Chapter 7	Conclusion	**27**
References	IEEE style; 10–20 references	**28-19**
Appendix	CAD drawings, calculations, questionnaires, etc.	**30**

List of Figures

Figure No.	Figure Name	Page No.
1.1	General Objective	5
1.2	Future Expansion	6
1.3	Final Assembly Charger	7
2.1	Novelty	12
3.1	Block Diagram	13
3.2	Flow Chart	14
3.3	Housing	15
3.4	Base	15
3.5	Cap	15
3.6	Light Guide	15
3.7	PCB Assembly	15
3.8	Sleeve	15
3.9	Cable Assembly	16
3.10	PCB Assembly	16
3.11	TTP	16
3.12	PCB ASSEMBLY	17
3.13	PCB LAYOUTING	17
4.1	Work Breakdown	19

List of Tables

Figure No.	Figure Name	Page No.
2.1	Literature Review	8-11
3.1	Component Material	18
4.1	Gannt Chart	20
4.2	Risk Assessment	20-21
5.1	Budget Assessment	22-23
5.2	BOM	23
6.1	SDG Mapping	25-26

Abbreviations and symbols

- USB: Universal Serial Bus
- USB-A: Universal Serial Bus Type-A
- USB-C: Universal Serial Bus Type-C
- PD: Power Delivery
- QC: Quick Charge
- DC: Direct Current
- DC-DC: Direct Current to Direct Current Converter
- IC: Integrated Circuit
- PCB: Printed Circuit Board
- ECU: Electronic Control Unit
- EMI: Electromagnetic Interference
- EMC: Electromagnetic Compatibility
- OVP: Over-Voltage Protection
- OCP: Over-Current Protection
- OTP: Over-Temperature Protection
- SCP: Short-Circuit Protection
- LED: Light-Emitting Diode
- GND: Ground
- VIN: Input Voltage
- VOUT: Output Voltage
- A: Ampere
- V: Volt
- W: Watt
- Ω: Ohm
- A+C: USB Type-A and USB Type-C
- CAN: Controller Area Network
- CAD: Computer-Aided Design
- BOM: Bill of Materials
- VP: Design Verification Plan
- DFMEA: Design Failure Mode and Effects Analysis

CHAPTER 1: INTRODUCTION

1.1 Background

With the increasing use of smartphones, tablets, GPS devices, smartwatches, and other portable electronic devices, the need for reliable charging facilities inside automobiles has become essential. Modern vehicles require convenient and safe charging solutions that can support different types of electronic devices.

An **Automotive USB A+C Charger** is an electronic power supply system designed to convert the vehicle's electrical supply into regulated USB charging power. The proposed charger incorporates both **USB Type-A and USB Type-C ports**, allowing compatibility with conventional as well as modern electronic devices.

The project focuses on the design and development of a compact automotive charger with suitable power conversion, voltage regulation, protection, and thermal management. The charger is intended to provide stable and reliable power while operating under the varying electrical conditions commonly found in automotive environments.

1.2 Problem Statement

Existing Problem

Many vehicles are equipped with limited charging ports, and older charging solutions may support only USB Type-A interfaces. The increasing adoption of USB Type-C devices creates a need for chargers that support both USB Type-A and USB Type-C connections.

Industrial Issue

Automotive electronic systems operate under varying input voltage conditions and may experience electrical disturbances such as voltage fluctuations, over-current conditions, short circuits, and temperature variations. Therefore, an automotive charger must provide regulated output power and adequate protection.

Research Gap

Existing low-cost charging solutions may not provide an optimum combination of **dual-port compatibility, compact design, protection features, reliability, and efficient power conversion**. There is a need for an improved automotive charging solution capable of supporting different devices while maintaining safe and stable operation.

Need for Improvement

A reliable Automotive USB A+C Charger is required to provide convenient charging for multiple devices while ensuring proper voltage regulation, protection against electrical faults, efficient operation, and suitability for automotive applications.

1.3 Need of the Project

The need for an Automotive USB A+C Charger arises from the growing number of USB-powered electronic devices used by vehicle occupants. A vehicle may need to charge multiple devices simultaneously, making dual-port charging highly useful.

The project is important because it aims to:

- Provide both USB Type-A and USB Type-C charging interfaces.
- Support charging of different generations of electronic devices.
- Provide regulated and stable output power.
- Improve user convenience by allowing multiple devices to be charged.
- Incorporate protection features for safe operation.
- Develop a compact and reliable charging solution suitable for automotive applications.
- Provide a practical solution that can be further developed for commercial automotive products.

1.4 Objectives

General Objective

To **design, develop, and test an Automotive USB A+C Charger** capable of providing safe, stable, and reliable charging power for multiple USB-powered electronic devices.

Specific Objectives

1. **Design** an automotive charging circuit capable of converting the vehicle input supply into regulated USB output power.
2. **Develop** a dual-port charging system incorporating both USB Type-A and USB Type-C interfaces.
3. **Implement** suitable electrical protection features such as over-voltage, over-current, short-circuit, and thermal protection.
4. **Optimize** the charger for efficient power conversion, compact size, and reliable operation under automotive conditions.
5. **Fabricate** and assemble a functional prototype of the Automotive USB A+C Charger.
6. **Test and validate** the prototype for output voltage, charging performance, protection functions, thermal performance, and overall reliability.

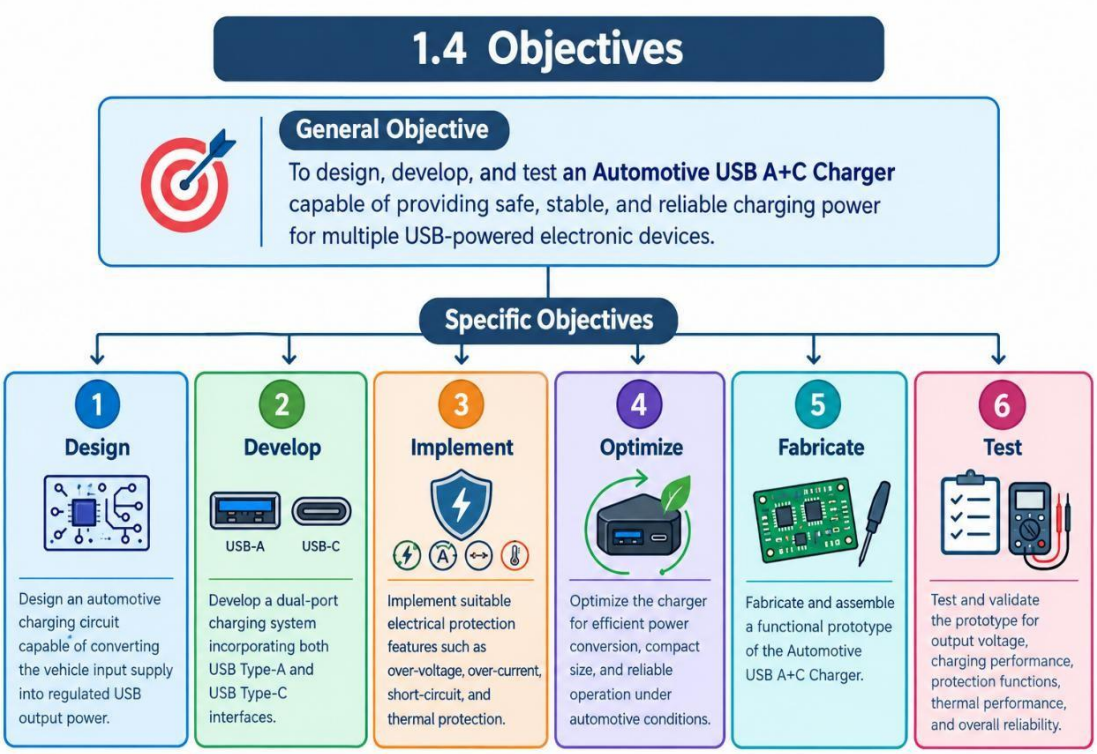


Fig 1.1: General Objectives

1.5 Scope

The scope of the project includes the design, development, fabrication, and testing of an Automotive USB A+C Charger.

The project will cover:

- Design of the automotive power-input section.
- DC-DC power conversion and voltage regulation.
- Integration of USB Type-A and USB Type-C charging ports.
- Selection of suitable electronic components.
- Protection against over-voltage, over-current, short circuit, and excessive temperature.
- PCB design and hardware development.
- Prototype fabrication and assembly.
- Electrical and functional testing of the charger.
- Evaluation of output voltage, current, efficiency, temperature, and charging performance.

Limitations

- The charger is primarily intended for automotive applications and depends on the vehicle's electrical supply.
- Charging performance depends on the supported power rating and charging requirements of the connected device.
- The project prototype may not include every advanced charging protocol available in commercial chargers. 14
- Long-term field validation under all possible automotive environmental conditions may be outside the scope of the academic project.

Future Expansion

The project can be further developed by incorporating:

- Higher USB-C Power Delivery (PD) capabilities.
- Support for additional fast-charging protocols.
- Multiple USB-C ports.
- Intelligent power-sharing between ports.
- Digital monitoring of voltage, current, and temperature.
- CAN-based vehicle communication and diagnostics.
- Improved thermal management.
- Automotive-grade EMC/EMI optimization.
- Advanced protection and fault-detection features.

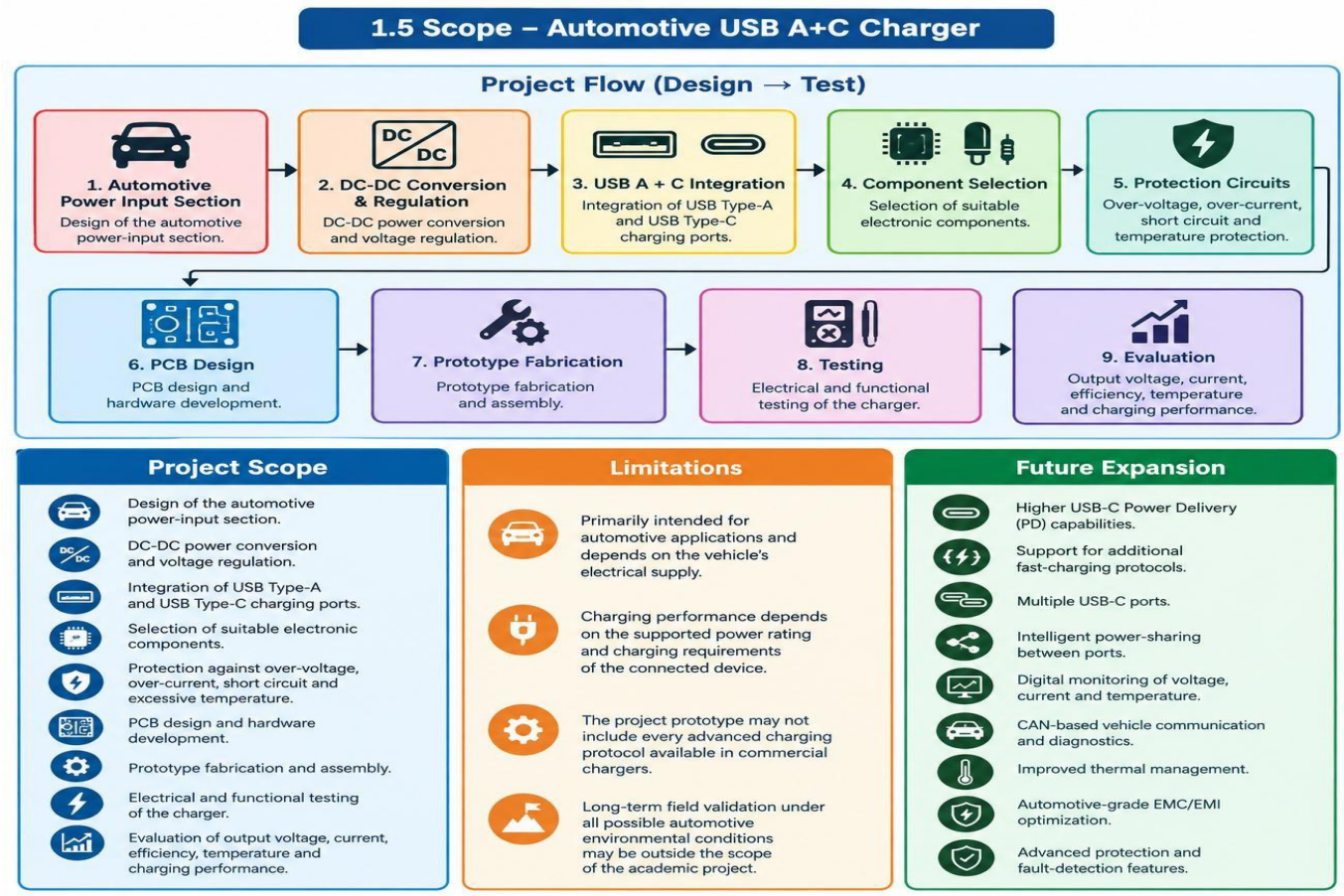


Fig 1.2: Future Expansion

1.6 Expected Outcomes

The expected outcomes of the project are:

- A **working prototype** of the Automotive USB A+C Charger.
- A functional USB Type-A and USB Type-C dual-port charging system.
- Stable and regulated charging output.
- Proper implementation of electrical protection features.

- A **PCB design and fabricated hardware prototype**.
- Successful testing and validation of the charger under defined operating conditions.
- Performance data related to voltage, current, efficiency, and temperature.
- Technical documentation including circuit diagrams, PCB design, testing results, and project report.
- A design that can potentially be further developed into a **commercial automotive charging product**.



Fig 1.3: Final Assembly of Charger

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The development of automotive charging systems has increased significantly with the growing use of smartphones, tablets, navigation devices, and other portable electronic devices inside vehicles. Modern automotive charging systems require efficient DC-DC conversion, regulated output voltage, protection against electrical faults, low electromagnetic interference, and compact packaging.

Previous research has mainly focused on improving **DC-DC converter efficiency, power density, current control, thermal performance, and protection mechanisms**. At the same time, industrial solutions have addressed automotive USB charging challenges such as cable voltage drop, USB charging identification, electromagnetic interference, and higher-power USB-C Power Delivery (PD).

2.2 Literature Review Table

Paper / Study	Year	Method / Approach	Major Findings	Research Gap / Limitation
Automotive DC-DC Converter	2022	A conventional DC-DC buck converter topology was investigated for converting the vehicle's higher DC battery voltage into a regulated low-voltage output suitable for electronic loads. The study considered switching control, current regulation, voltage feedback, and basic protection mechanisms.	The study demonstrated that a properly controlled buck converter can provide stable output voltage under variations in input voltage and load current. Current control also improves converter protection during overload and short circuit conditions.	The study mainly focused on the DC-DC conversion stage and did not address the complete integration of an automotive USB charging system. In particular, USB-A and USB-C dual-port operation, connector limitations, USB-C PD requirements, compact packaging, and automotive EMI/EMC requirements were not comprehensively addressed.
EV DC-DC Converter Review	2022	Different DC-DC converter topologies used in electric-vehicle applications were compared based on efficiency, switching losses, voltage conversion ratio, size, and power-handling capability.	The review identified several converter architectures capable of achieving high conversion efficiency. It also showed that topology selection depends strongly on required output power, input-voltage range, efficiency, switching frequency, and thermal performance.	The primary focus was on EV powertrain and charging applications, which generally operate at significantly higher power levels than automotive USB charging systems. Therefore, the findings do not directly address the requirements of a compact low/medium-power USB charger with multiple output ports.

Paper / Study	Year	Method / Approach	Major Findings	Research Gap / Limitation
Automotive USB Power	2022	The study examined automotive USB power-management requirements, including input-voltage variation, voltage regulation, current limitation, protection against abnormal conditions, and electromagnetic interference.	The study highlighted that automotive USB chargers must maintain a stable output despite variations in the vehicle electrical system. It also emphasized the importance of over-voltage protection (OVP), over-current protection (OCP), short-circuit protection, thermal protection, and EMI control.	A complete dual-port USB-A + USB-C automotive charger was not comprehensively developed. The interaction between the two ports during simultaneous charging, power allocation, PCB layout, thermal management, and mechanical integration remained insufficiently explored.
DC-DC Converter for EV Charging	2023	High-efficiency DC-DC conversion techniques were investigated for automotive and EV charging applications. The study focused on reducing switching and conduction losses and improving overall conversion efficiency.	Advanced switching techniques and optimized converter designs can significantly improve power-conversion efficiency. Reduced losses also contribute to lower thermal generation and improved system reliability.	The proposed concepts were primarily intended for higher-power EV charging systems. Such architectures can be unnecessarily complex, expensive, and physically large for a compact automotive USB charger. Their suitability for low-power USB-A/USB-C applications was therefore not fully established.
Multi-Output DC-DC Converter	2023	A multi-output DC-DC converter architecture was investigated to provide regulated power to more than one output simultaneously. The study considered output regulation and power distribution between multiple loads.	Multi-output architectures can provide simultaneous power delivery while reducing the requirement for separate conversion stages. Proper control of the outputs can help maintain voltage stability when the load on one output changes.	The converter was not specifically designed for USB charging applications. USB charging standards, USB-C Power Delivery negotiation, USB-A charging protocols, connector heating, automotive packaging, and port-specific protection were not the main focus.
Gan-Based Automotive Converter	2024	A Gallium Nitride (Gan)-based high-frequency converter was investigated to increase	Gan switching devices can operate at higher frequencies with lower switching losses	Gan-based designs generally introduce increased circuit complexity and

Paper / Study	Year	Method / Approach	Major Findings	Research Gap / Limitation
		switching frequency and reduce the size of passive components. The study evaluated efficiency, power density, switching performance, and thermal behaviour.	than many conventional silicon-based devices. This can result in improved power density, smaller magnetics, reduced converter size, and potentially higher efficiency.	component cost. Gate-drive design, PCB layout, thermal management, EMI control, and automotive qualification can also be more challenging. Therefore, the technology may not always be economically justified for a relatively low-cost automotive USB charger.
Automotive USB-C Charger	2024	A buck-boost based power-conversion approach was investigated for automotive USB-C charging. The converter was designed to accommodate variations in automotive input voltage while providing a regulated USB-C output suitable for higher charging power.	The study demonstrated that buck-boost conversion can provide a stable USB-C output over a wider input-voltage range. The approach is particularly useful when the automotive supply voltage varies significantly and higher USB-C charging power is required.	The work primarily concentrated on USB-C charging and provided limited investigation of a combined USB-A + USB-C solution. Simultaneous charging, power-sharing between ports, compact PCB design, thermal interaction, and automotive mechanical integration require further investigation.
EV Charging Converter	2024	A bidirectional DC-DC converter was studied for EV charging applications. The converter enabled controlled power flow between the vehicle battery and external electrical systems. Efficiency, power regulation, and bidirectional operation were evaluated.	Bidirectional converters provide flexible power-flow control and can achieve high efficiency in EV charging applications. They are useful for applications such as vehicle-to-grid and vehicle-to-load systems.	Bidirectional operation is unnecessary for conventional USB charging. Such converters introduce additional switches, control circuits, sensing requirements, and cost. Therefore, they are generally not suitable for a compact, low-power automotive USB charger where simplicity, reliability, and cost are important.
Automotive Charging Technology Review	2025	A broad review of automotive charging architectures, DC-DC conversion techniques, charging interfaces, power-management approaches, and system-	The review identified a trend toward compact, efficient, high-power-density charging systems. It also emphasized the importance of thermal	The literature continues to emphasize vehicle-level and EV charging systems rather than compact auxiliary charging modules. A detailed investigation combining a

Paper / Study	Year	Method / Approach	Major Findings	Research Gap / Limitation
		level requirements was carried out.	management, protection, power density, efficiency, and integration within modern vehicles.	conventional 12-V automotive supply, DC-DC conversion, USB-A, USB-C, simultaneous charging, protection, EMI/EMC, thermal considerations, and compact mechanical packaging remains limited.
USB-C Power Delivery Technology	2025	USB-C Power Delivery (PD) based power-management techniques were studied, including voltage/current negotiation, power-profile selection, controlled output regulation, and communication between the source and connected device.	USB-C PD enables significantly greater charging flexibility compared with conventional fixed-voltage USB charging. Dynamic power negotiation allows the charger to provide an appropriate voltage and current according to the connected device's requirements.	USB-C PD requires additional control and communication circuitry, increasing system complexity. Most studies focus on USB-C as a standalone charging interface and do not fully address integration with a USB-A port in an automotive environment, especially with simultaneous charging, power budgeting, EMI/EMC, thermal limitations, and compact packaging.

Table 2.1: Literature Review

2.3 Research Gap

Existing research mainly focuses on **DC-DC converters and EV charging systems**. Commercial automotive USB chargers are available, but there is a need for a **compact, reliable, and cost-effective system combining USB Type-A and USB Type-C**.
The proposed project addresses this gap by integrating:

- Automotive DC-DC power conversion
- USB Type-A and Type-C ports
- Voltage regulation
- Over-current and short-circuit protection
- Compact PCB design
- Reliable multi-device charging

2.4 Novelty / Innovation

The main innovation of the project is the **integration of USB Type-A and USB Type-C charging in a single automotive charger**.
Key features include:

- Dual USB charging interfaces
- Compact design
- Regulated output
- Protection against electrical faults
- Efficient power conversion
- Support for simultaneous charging
- Scope for future USB-C fast charging/PD integration



Fig 2.1: Novelty version of Charger

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

The proposed system is an **Automotive USB A+C Charger** designed to convert the vehicle's DC supply into regulated charging power for electronic devices. The system provides both **USB Type-A and USB Type-C outputs**.
The main sections of the system are the automotive input, protection circuit, DC-DC converter, control and regulation circuit, and USB output ports. The system is designed to provide stable output voltage while protecting the charger and connected devices from electrical faults.

3.2 Working Principle

The working of the proposed charger is as follows:

1. The vehicle battery provides the **DC input supply** to the charger.
2. The input protection circuit protects the system from electrical disturbances.
3. The DC-DC converter converts the input voltage into the required regulated voltage.
4. The control circuit monitors and regulates the output.
5. The regulated power is supplied to the **USB Type-A and USB Type-C ports**.
6. Connected devices receive the required charging power.
7. Protection circuits operate in case of **over-current, short circuit, over-voltage, or excessive temperature**.

3.3 System Architecture / Block Diagram

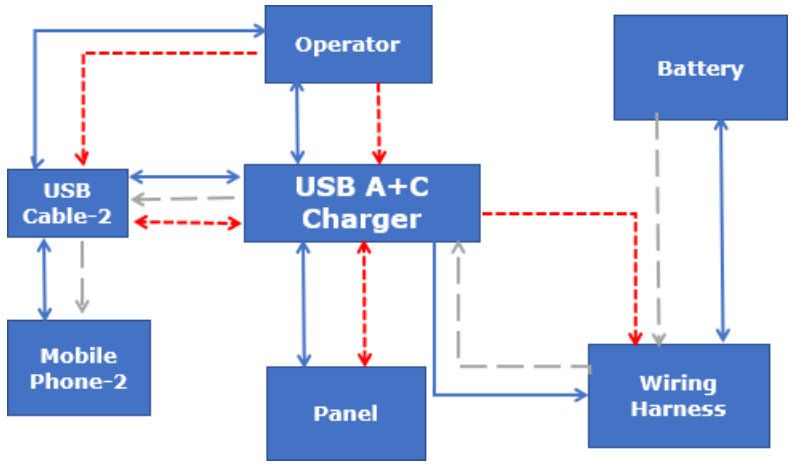


Fig 3.1: Block Diagram/System Architecture

3.4 Flow Chart

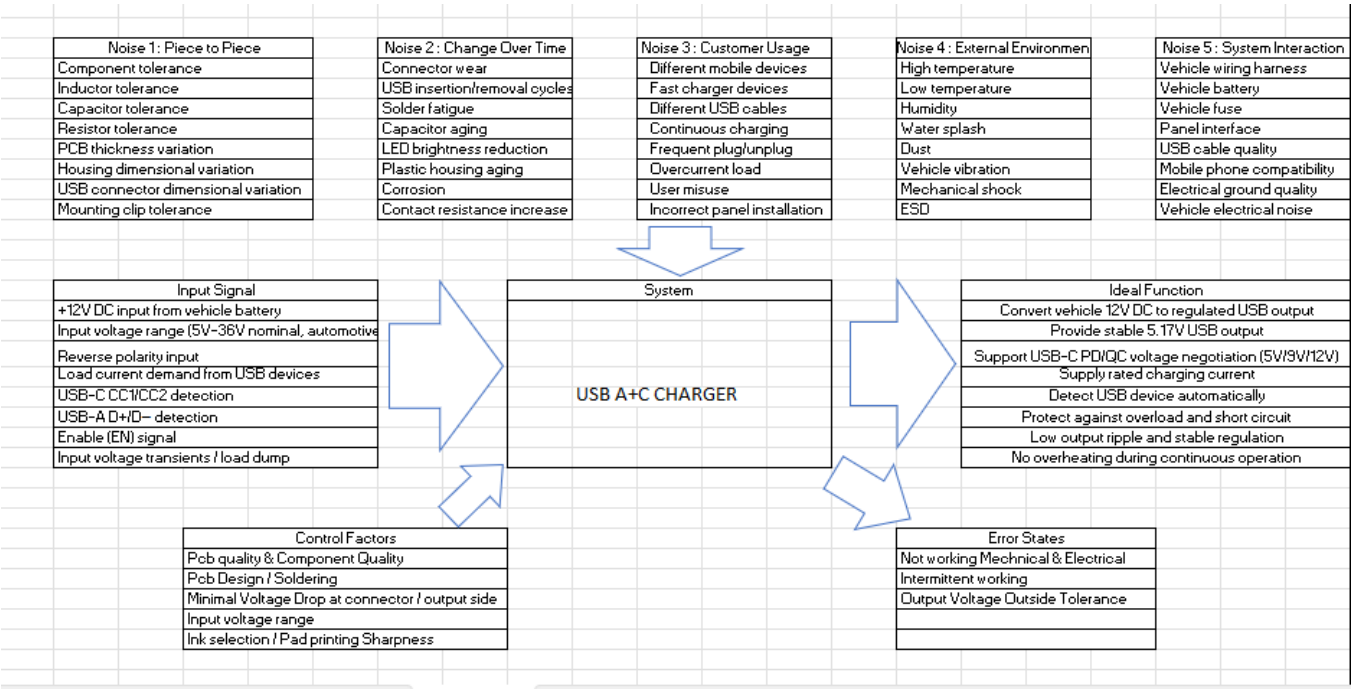


Fig 3.2: Flow chart/Process of Working

3.5 Methodology

Design Procedure

- Identify the required automotive input and USB output specifications.
- Select a suitable DC-DC converter topology.
- Select suitable protection components.
- Design the circuit diagram.
- Calculate required component ratings.
- Design the PCB.
- Verify the design through simulation/calculation where applicable.



HOUSING
Material Used – PBT
GF 30%

Fig 3.3: Housing



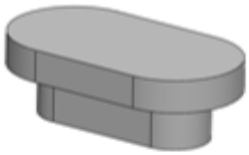
BASE
Material Used – PBT
GF 30%

Fig 3.4: Base



CAP
Material Used –
TPE(GRADE-1160E)

Fig 3.5: Cap



LIGHT GUID
Material Used – PC
TRANSPARENT + 6%
ROMA WHITE

Fig 3.6: Light Guid



PCB ASSEMBLY

Fig 3.7:PCB Assembly



Sleeve
Material Used – PVC

Fig 3.8: Sleeve





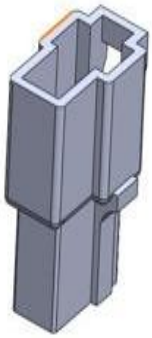


Fig 3.9: Cable Assembly **Fig 3.10: Terminal** **Fig 3.11: TTP**

Experimental Procedure

- Assemble the prototype PCB.
- Apply the automotive DC input.
- Measure the output voltage and current.
- Test USB-A and USB-C charging outputs.
- Check operation under different load conditions.
- Test protection functions.
- Measure temperature and efficiency.
- Record and analyse the results.

Manufacturing Process

The manufacturing process consists of:
Circuit Design → PCB Design → PCB Fabrication → Component Assembly → Soldering → Inspection → Testing → Final Assembly

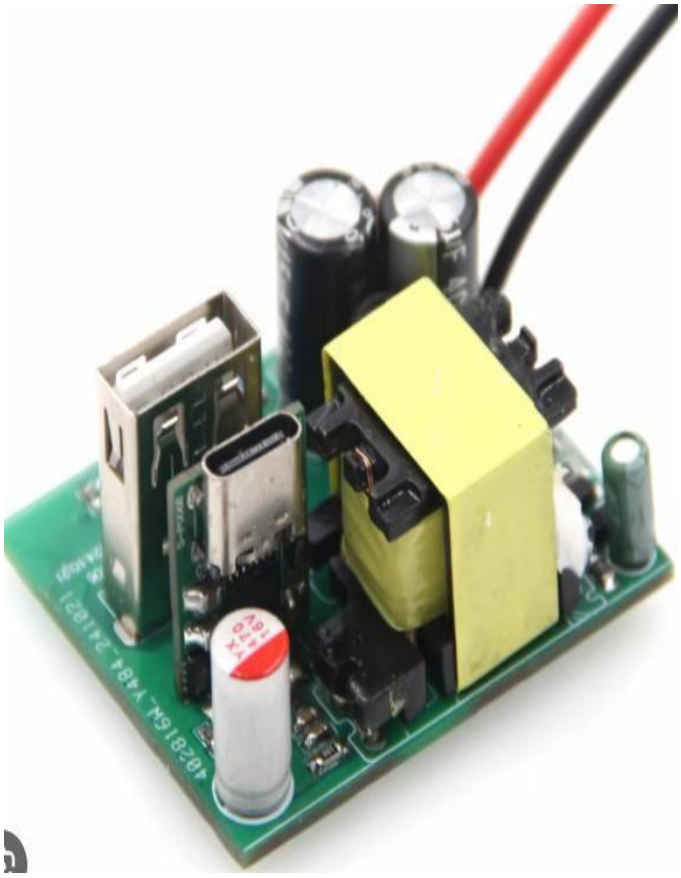


Fig 3.12: PCB Assembly

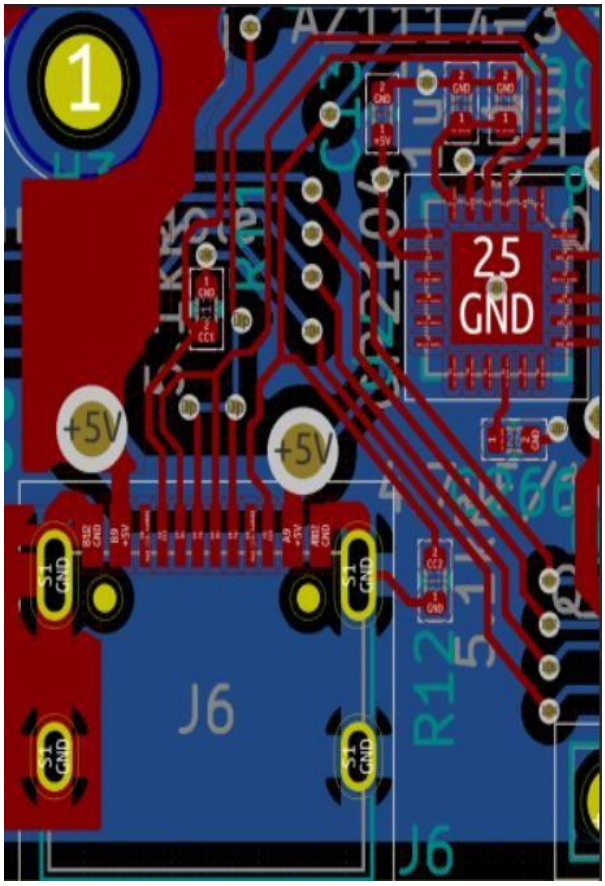


Fig 3.13: PCB Lay outing

Programming

Programmable controller is used; the programming procedure includes:

1. Initialize the microcontroller.
2. Monitor input/output parameters.
3. Monitor fault conditions.
4. Control the power-conversion section.
5. Implement protection logic.
6. Test and debug the firmware.

3.6 Components / Materials

3.6 Components / Materials

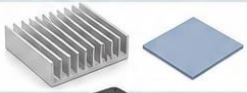
Component	Image	Specification	Quantity
DC-DC Converter IC		Automotive-rated, required output voltage/current	1
USB Type-A Connector		Standard USB-A charging connector	1
USB Type-C Connector		USB-C charging connector	1
Input Protection Fuse		Automotive-rated fuse	1
TVS Diode		Automotive transient protection	1
Inductor		As per DC-DC converter design	1
Capacitors		Input/output filtering	As required
Resistors		Feedback and protection circuit	As required
MOSFET		Power switching, suitable voltage/current rating	As required
PCB		FR-4, designed for charger circuit	1
Heat Sink / Thermal Pad		As required for thermal management	As required
Enclosure		Automotive-grade housing	1

Table 3.1: Components/Materials

CHAPTER 4: PROJECT PLANNING

4.1Work Breakdown Structure

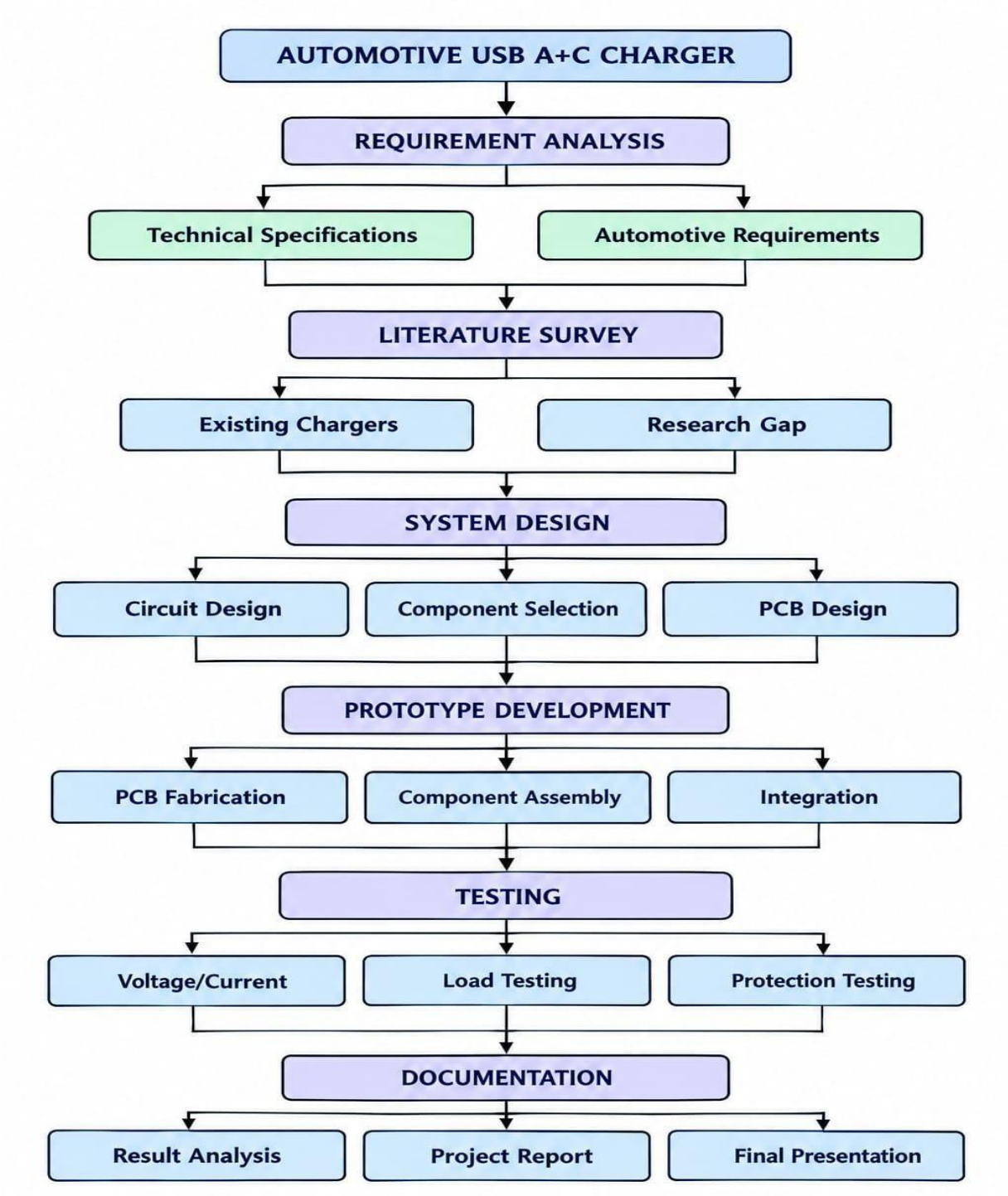


Fig 4.1: Work Breakdown Structure

4.2 Gantt Chart

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Requirement Analysis	✓					
Literature Review	✓	✓				
System Design		✓	✓			
Component Selection		✓	✓			
Circuit & PCB Design			✓	✓		
Prototype Development				✓	✓	
Testing & Validation					✓	✓
Result Analysis					✓	✓
Report Preparation						✓
Final Presentation						✓

Table 4.1: Gannt Chart

4.3 Roles and Responsibilities

Student Responsibility

Shubham Chaugule : Project coordination, system design and documentation

Manish Devashi : Circuit design, component selection and hardware development

Halim Ansari : PCB design, assembly and testing

Mushkann Das : Research, testing, result analysis and documentation

4.4 Risk Assessment

Risk	Impact	Mitigation
Incorrect input voltage	Circuit damage	Use proper input protection and voltage-rated components
Over-current	Component overheating/damage	Implement current limiting and suitable fuse
Short circuit	System failure	Provide short-circuit protection
Excessive temperature	Reduced reliability	Use thermal management and proper PCB layout
DC-DC converter failure	No charging output	Select components with suitable voltage/current ratings
PCB design error	Prototype malfunction	Perform schematic and PCB design verification before fabrication
USB connector damage	Charging interruption	Use suitable automotive-grade connectors
Output voltage fluctuation	Device charging problems	Use proper voltage regulation and filtering
Component availability	Project delay	Select alternative equivalent components in advance

Risk	Impact	Mitigation
Testing failure	Project delay	Perform step-by-step testing during prototype development

Table 4.2: Risk Assessment

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The proposed Automotive USB A+C Charger is technically feasible because the required electronic components and technologies are readily available. The system uses commonly available DC-DC converter components, USB connectors, protection devices, capacitors, resistors, and a PCB.

The charger can be designed, assembled, and tested using standard electronic design and testing tools. The system can be evaluated for output voltage, output current, efficiency, temperature, and protection performance.

5.2 Economic Feasibility

The project is economically feasible because it uses commercially available and relatively low-cost electronic components. The prototype can be developed at a reasonable cost compared with commercially available automotive charging modules.

The main expenses include the DC-DC converter, USB connectors, PCB, protection components, passive components, and enclosure.

The design can also be optimized for mass production by reducing component count and selecting cost-effective components.

5.3 Operational Feasibility

The proposed charger is easy to operate. It can be connected to the vehicle's DC power supply and used to charge compatible electronic devices through the USB Type-A or USB Type-C port.

The system is designed to operate with minimum user interaction. Protection features help improve safe operation during abnormal conditions such as over-current, short circuit, or excessive temperature.

5.4 Budget Estimation

The estimated prototype cost is shown below:

Item	Quantity	Estimated Cost (₹)
DC-DC Converter IC/Module	1	150
USB Type-A Connector	1	30
USB Type-C Connector	1	50
MOSFET / Switching Components	2	80
Inductor	1	50
Capacitors	6	60
Resistors	10	30

Item	Quantity	Estimated Cost (₹)
TVS Diode / Protection Components	2	60
Fuse & Holder	1	30
PCB Fabrication	1	300
Heat Sink / Thermal Pad	1	80
Wires & Connectors	As required	100
Enclosure	1	150
Total Estimated Cost		**₹1,170**

Table 5.1: Budget Assessment

5.5 Bill of Materials (BOM)

Bill of Materials (BOM) – Automotive USB A+C Charger

Sr. No.	Component	Image	Specification	Quantity
1	DC-DC Converter IC		Automotive-rated, suitable input/output rating	1
2	USB Type-A Connector		Standard USB-A charging port	1
3	USB Type-C Connector		USB-C charging port	1
4	MOSFET		Suitable voltage and current rating	2
5	Inductor		As per converter design	1
6	Capacitor		Input/output filtering	6
7	Resistor		Feedback/current sensing	10
8	TVS Diode		Automotive transient protection	2
9	Fuse		Automotive-rated	1
10	PCB		FR-4 PCB	1
11	Heat Sink/Thermal Pad		Suitable for power components	1
12	Wires & Connectors		Automotive suitable	As required
13	Enclosure		Compact automotive housing	1

Table 5.2: Bills of Materials

CHAPTER 6:
EXPECTED RESULTS & IMPACTS

The proposed **Automotive USB A+C Charger** is expected to provide reliable and stable charging performance for compatible electronic devices. Since this section is part of the project synopsis, only the **expected results** are presented and no actual test results are included.

6.1 Expected Prototype

A functional prototype of the Automotive USB A+C Charger is expected to be developed with **USB Type-A and USB Type-C charging ports**. The prototype should be compact, reliable, and suitable for automotive power supply conditions.

6.2 Expected Accuracy

The charger is expected to maintain the required output voltage within the specified tolerance under different load conditions. The protection and monitoring circuits are also expected to respond correctly to abnormal operating conditions.

6.3 Expected Efficiency

The DC-DC conversion stage is expected to provide **good power-conversion efficiency** with minimum power loss. Proper component selection and PCB design are expected to help reduce switching and conduction losses.

6.4 Expected Performance

The system is expected to:

- Provide stable charging output.
- Support USB Type-A and USB Type-C devices.
- Operate reliably with automotive DC input.
- Provide protection against over-current and short circuit.
- Maintain acceptable operating temperature.
- Support charging of multiple devices within the designed power limit.

6.5 Expected Simulation Results

Circuit simulation is expected to verify:

- Input and output voltage levels.
- Converter switching operation.
- Output voltage regulation.
- Load response.
- Current flow under different load conditions.
- Expected power losses and efficiency.

6.6 Expected Experimental Results

After prototype fabrication, testing is expected to demonstrate:

- Proper operation of USB Type-A and USB Type-C outputs.
- Stable output voltage under different loads.
- Expected output current capability.

- Acceptable efficiency.
- Proper operation of protection features.

- Acceptable thermal performance.
- Successful charging of compatible USB devices.

6.7 Industrial Relevance

The proposed **Automotive USB A+C Charger** has good industrial relevance due to the increasing demand for USB charging facilities in modern vehicles. The system can be further developed for use in **passenger cars, commercial vehicles, buses, and other automotive applications**.

The compact design, dual USB interfaces, power regulation, and protection features make the project suitable for further development into a commercial automotive charging module.

6.8 Societal Impact

The project provides a convenient method for charging electronic devices while travelling. The availability of both **USB Type-A and USB Type-C** ports improves compatibility with different devices.

It can also improve user convenience by allowing multiple compatible devices to be charged from a single automotive charging unit.

6.9 Environmental Impact

The project focuses on efficient power conversion to reduce unnecessary energy losses during charging. Proper component selection and thermal management can improve the overall energy efficiency and operating life of the product.

The design can also support sustainable manufacturing by using a **compact PCB, optimized component selection, and recyclable materials where practical**. Future versions can further improve efficiency and reduce electronic waste through longer product life and modular design

6.10 SDG Mapping

The project contributes to the following **United Nations Sustainable Development Goals (SDGs)**:

SDG	Goal	Project Contribution
☑ **SDG 7**	Affordable and Clean Energy	Promotes efficient electrical power conversion and reduced energy losses.
☑ **SDG 9**	Industry, Innovation and Infrastructure	Develops an innovative automotive electronic charging solution.
☑ **SDG 11**	Sustainable Cities and Communities	Supports modern transportation by providing convenient in-vehicle charging.

☑ **SDG 12**	Responsible Consumption and Production	Encourages efficient component utilization and longer product life.

SDG	Goal	Project Contribution
☑ **SDG 13**	Climate Action	Improved energy efficiency can help reduce unnecessary energy consumption.

Table 6.1: SDG Mapping

CHAPTER 7: CONCLUSION

The **Automotive USB A+C Charger** project aims to address the increasing need for a reliable and convenient charging solution in modern vehicles. Existing charging systems may provide limited connectivity or support only a single type of USB interface.

The proposed system provides a **dual-port USB Type-A and USB Type-C charging solution** using an automotive DC-DC power conversion stage. The system is designed to provide regulated output power along with suitable protection against electrical faults such as over-current, short circuit, and over-voltage.

The expected contribution of this project is the development of a **compact, efficient, reliable, and practical automotive charging prototype**. The project can serve as a foundation for future improvements such as higher-power USB-C Power Delivery, intelligent power sharing, improved thermal management, and advanced automotive protection features.

The design and development process will involve selecting suitable components, preparing the circuit and PCB layouts, assembling the prototype, and carrying out systematic testing. The input stage, conversion circuit, output regulation, and protection functions will be evaluated to confirm that the charger operates as intended. Measurements of output voltage, current, efficiency, and temperature will help assess its performance under defined operating conditions.

Providing both USB Type-A and USB Type-C interfaces in one compact unit will improve compatibility with a broad range of existing and newer electronic devices. The design also considers the practical demands of an automotive environment, where the input supply may fluctuate and components can be exposed to heat and electrical disturbances. Appropriate component ratings, circuit protection, PCB layout, and thermal management are therefore important to achieving safe and dependable operation.

Although the project synopsis describes expected performance rather than completed test results, the proposed prototype will provide a basis for practical verification and further refinement. Testing will identify any limitations in regulation, charging performance, thermal behaviour, or protection response, allowing improvements to be made before a more advanced version is developed. The final evaluation will be based on measured results and the extent to which the prototype meets its design objectives.

The project also emphasizes the importance of balancing performance, safety, size, and cost during product development. Careful testing of the prototype will help confirm whether the selected components and circuit arrangement satisfy the intended requirements. Any issues observed during validation can guide changes to component selection, PCB routing, protection design, or thermal management. Documenting these observations will make the design process more systematic and provide useful information for future development. The work therefore combines theoretical knowledge of power electronics with practical design, assembly, and testing activities, while keeping the final solution focused on the charging needs of vehicle users.

In conclusion, the Automotive USB A+C Charger is a practical engineering project that combines power conversion, voltage regulation, dual-port connectivity, and electrical protection in a compact automotive application. With successful implementation and validation, it can offer a useful foundation for future enhancements such as USB-C Power Delivery, intelligent power sharing, improved efficiency, and more comprehensive automotive-grade protection. These developments could help make the system more capable and suitable for wider real-world use.

REFERENCES

A. Research Papers / Technical Literature

Use IEEE numbering like this:

[1] J. Liu *et al.*, "Deep integrated development of on-board charger and DC/DC on component level for new energy vehicles," *SAE Technical Paper*, 2022.

[2] V. S. Vijayakumar and N. Sudhakar, "A review on unidirectional converters for on-board chargers in electric vehicle," *Frontiers in Energy Research*, vol. 10, 2022.

[3] G. Kanimozhi *et al.*, "An effective charger for plug-in hybrid electric vehicles with enhanced PFC rectifier and ZVS-ZCS DC/DC converter," *International Journal of Energy Research*, 2022.

[4] G. Ramanathan *et al.*, "Design of single-inductor double-input double-output DC-DC converter for electric vehicle charger," *Materials Today: Proceedings*, 2022.

[5] S. B. Sudarshan and G. Arunkumar, "Isolated DC-DC power converters for simultaneous charging of electric vehicle batteries," *Sustainability*, vol. 15, no. 3, 2023.

[6] "Recent developments in DC-DC converter topologies for light electric vehicle charging: A critical review," *Applied Sciences*, vol. 13, no. 3, 2023.

[7] Snehalika, R. Patel, and C. K. Panigrahi, "GaN-based isolated bidirectional multiport DC-DC converter for electric vehicle charging," 2024.

[8] "A bidirectional DC/DC converter for renewable energy source-fed EV charging stations," *Results in Engineering*, 2024.

[9] "System-level compact review of on-board charging technologies for electrified vehicles," *Batteries*, vol. 11, no. 9, 2025.

[10] "A voltage-emulated peak current controlled buck converter for automotive applications with in-built over-current protection," *Microelectronics Journal*, 2022.

[11] "A valley-sensed emulated peak current controlled buck converter for automotive applications," *Microelectronics Journal*, 2022.

B. Standards

[12] USB Implementers Forum, *Universal Serial Bus Specification, Revision 2.0*, USB-IF.

[13] USB Implementers Forum, *USB Type-C Cable and Connector Specification*, USB-IF.

[14] USB Implementers Forum, *USB Power Delivery Specification, Revision 3.2*, USB-IF, 2026. The current USB-IF document library lists USB PD Revision 3.2 Version 1.2. (USB-IF)

[15] USB Implementers Forum, *USB Type-C and Power Delivery Source Power Requirements Test Specification*, USB-IF. (USB-IF)

[16] USB Implementers Forum, *USB Power Delivery Compliance Test Specification*, USB-IF, Rev. Q3 2026. (USB-IF)

C. Automotive Standards

[17] International Organization for Standardization, *ISO 16750: Road Vehicles — Environmental Conditions and Testing for Electrical and Electronic Equipment*, ISO.

[18] International Organization for Standardization, *ISO 7637: Road Vehicles — Electrical Disturbances from Conduction and Coupling*, ISO.

[19] International Organization for Standardization, *ISO 11452: Road Vehicles — Component Test Methods for Electrical Disturbances from Narrowband Radiated Electromagnetic Energy*, ISO.

D. Books

[20] R. W. Erickson and D. Maksimović, *Fundamentals of Power Electronics*, 3rd ed. Cham, Switzerland: Springer, 2020.

[21] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed. Hoboken, NJ, USA: Wiley.

[22] M. H. Rashid, *Power Electronics: Circuits, Devices, and Applications*, 4th ed. Pearson.

E. Websites / Technical References

[23] Analog Devices, "Automotive USB Power: Managing Increasing Currents, Longer Cables and Portable-Device

Specifications."
[24] Toshiba Electronic Devices & Storage Corporation, "Automotive USB Power Charger."
[25] USB Implementers Forum, "USB Power Delivery," USB-IF. The USB-IF document library provides the current USB Type-C, USB PD, and compliance specifications. (USB-IF

APPENDICES

The appendices contain supporting technical documents and design information related to the **Automotive USB A+C Charger** project. These documents provide additional details that support the design, development, and testing of the proposed system.

Appendix A – Circuit Diagram
The complete circuit schematic of the Automotive USB A+C Charger, including the automotive input stage, protection circuit, DC-DC converter, regulation circuit, and USB Type-A and Type-C output sections.

Appendix B – PCB Layout
The PCB layout showing the placement and routing of major components, power tracks, ground connections, and USB connectors.

Appendix C – CAD / Mechanical Drawing

The mechanical drawing of the charger enclosure, including overall dimensions, mounting arrangement, USB port openings, and component placement.

Appendix D – Bill of Materials

Detailed BOM containing component names, part numbers, specifications, quantities, and estimated costs.

Appendix E – Testing Data
Test formats for recording:

- Input voltage
- Output voltage
- Output current
- Load condition
- Power consumption
- Efficiency
- Temperature
- Protection test results
-

Appendix F – Project Photographs
Photographs of the following stages may be included:

1. Component selection
2. PCB fabrication
3. PCB assembly
4. Prototype assembly
5. Testing setup
6. Final Automotive USB A+C Charger prototype

Appendix G – Project Canvas
A project canvas containing the **problem statement, proposed solution, target users, key features, resources, expected benefits, and future scope**.